

Supply Chain Risk & Demand Optimization Analysis

Business Case Study | Retail Analytics | Inventory Risk | Demand Planning

Project Overview

The retail firm had been facing challenges with frequent stockouts, lack of availability of inventory, and losses due to revenue leakage for popular items.

The project aims at examining the sales performance of the business, its inventory risks, customer buying patterns, and the efficacy of promotional activities to determine bottlenecks in the operations.

The key activities in the project include:

- Retail Sales Analytics
- Inventory Risk Analysis
- Supply Chain Performance Measurement
- Demand Trends Analysis
- Business Intelligence Dashboarding
- Strategic Recommendations Generation

Business Problem

Some of the operational problems encountered by retailers include:

- Frequent shortages of popular products
- Inefficient inventory allocation
- Poor seasonal demand forecasting
- Revenue leakages resulting from lack of stocks
- Limited visibility into stock and sales data

The organization sought answers to the following business queries:

- What are the top revenue-generating products and categories?
- Which products face the highest inventory risk?
- How does stockout impact the sales performance?
- Does promotion boost demand?
- How can operational efficiencies minimize revenue leakage?

Business Objective

This business project was intended to:

- Analyze retail sales and inventory performance
- Highlight high risk products that have been subject to inventory stockouts
- Evaluate the effect of inventory shortage on operations
- Enhance inventory planning and demand forecasting
- Inform supply chain decisions with data
- Propose measures to minimize financial losses

Tools & Technologies

Tool	Purpose
SQL	Business analysis & KPI reporting
Power BI	Dashboarding & visualization
Excel	Data cleaning & preprocessing

Dataset Overview

- Retail & supply chain transactional dataset
- 5000+ retail transactions

Key Business KPIs

KPI	Value
Total Revenue	₹15.26M
Total Transactions	5000
Total Quantity Sold	14.9K
Average Order Value	₹3052
Stockout Rate	51.86%

Business Analysis & Insights

1. Sales Performance Analysis

The analysis was performed based on:

- Monthly sales trend
- Contribution made by different products
- Category sales
- Monthly sales trend per store
- Purchase behavior of customers

Insights

- Electronics was the most revenue contributing category
- TVs and Tablets were found to be the most successful products
- Highest sales occurred in August, suggesting seasonal sales trends
- The loyal customers accounted for the major revenue contribution

Impact on Business

There was substantial demand for electronics products from the customers; however, poor logistics and supply chain process efficiency prevented the organization from capitalizing on the demand.

Recommendation

Inventory planning for electronics products during peak demand seasons.

2. Inventory Risks and Stockouts Analysis

The analysis considered:

- Frequency of stockouts for each product
- Inventory shortages at different locations
- Demand behavior during stockouts
- Supply risks in operations

Insights

- The stockout percentage was 51.86%, showing serious inventory management issues
- Tablets, Smartphones, and Fridges had the highest stockout percentages
- High-demand products had inventory shortages in their peak sales seasons
- The demand remained high regardless of product unavailability

Critical Business Insight

Some of the best-selling items earned considerable profit, but also showed frequent stockouts.

This pointed towards:

- Missed sale opportunities
- Revenue loss
- Inefficient inventory management
- Dissatisfied customers

Impact on Business

Stockouts negatively impacted the company's potential to make profits from high-demand products.

Recommendation

Adopt safety stock strategies and conduct accurate demand forecasting for risky products.

3. Revenue Leakage Analysis

The analysis uncovered some of the operational issues arising from stockouts and discrepancies between demand and supply.

Insights

- The high-demand products were out of stock at times of peak demand
- The business experienced poor revenue generation despite consistent customer demand growth after September
- The stockouts limited the firm's ability to process orders.

Impact on Business

The lack of stock impacted the operations of the business directly in that the company could not capitalize on the customer demand.

Recommendation

Inventory risk management should be established.

4. Promotion Effectiveness Analysis

Analysis focused on:

- Money made from promotional activities
- Type of promotion effectiveness
- Sales demand influence from the use of discounts

Insights

- Moderate increase in sales achieved through promotions
- Influence of product availability on revenue was greater than the influence of discounting practices
- Money-making abilities from percentage discounts and BOGO promotions were minimal

Impact on business

The business utilized promotions despite having difficulties in inventory availability.

Recommendation

Promotions should focus more on:

- Slow-moving inventory
- Profitable products
- Segment marketing to customers

Root Cause Analysis

The project identified multiple operational drivers behind declining sales performance after September.

Operational Issue	Business Impact
High stockout frequency	Lost sales opportunities
Weak demand forecasting	Inventory shortages during peak periods
Poor inventory allocation	Regional supply-demand mismatch
Overdependence on promotions	Limited revenue improvement

Estimated Business Impact

This study found that many threats to business operations arise from inventory shortages.

Operational Threats

- Loss of income as a result of inventory shortages
- Lack of satisfaction among customers
- Operational inefficiencies
- Poor inventory visibility in high-demand areas

Strategic Opportunities- Lowering the occurrence of stockouts and better demand planning may positively impact:

- Conversions of revenue
- Inventory turnover
- Customer retention
- Operational efficiency
- Supply chain stability

Recommendations

1. Implement Demand Forecasting

The organization must implement demand forecasting models that utilize past sales data and seasonality trends to forecast future demand for profitable products. The implementation of these models will prevent stockout problems during peak seasons, enhance inventory management, and promote revenue stability.

2. Increase Safety Stock for Riskier Items

The business must hold sufficient safety stocks for riskier items like TVs, Tablets, and Smartphones to avoid inventory gaps. This recommendation will decrease revenue losses, boost order fill rates, and enhance customer satisfaction.

3. Allocate Inventory Based on Regional Trends and Store Performance

The business must manage its inventory dynamically according to regional demand trends and store performance metrics to guarantee optimal inventory allocation across regions. The proposed allocation technique will optimize inventory turnover rates, eliminate regional stockout problems, and streamline operations.

4. Optimize Promotion Approach

The company should improve its promotional approach by focusing on slow-moving inventory, high-margin products, and personalized promotions for specific customers.

5. Develop a Dashboard for Inventory Risk Monitoring

An inventory risk monitoring dashboard is required to be developed by the business for tracking high stockout probability products, inventory level warning, seasonal demand increase, and regional inventory risks. This will help in improving the speed of action, enhancing supply chain visibility, and aiding in making decisions.

Dashboard Features

The Power BI dashboard includes:

- Revenue KPI tracking
- Monthly sales trends
- Product-wise revenue analysis
- Inventory risk visualization
- Promotion effectiveness analysis
- Geographic sales analysis
- Customer segmentation insights
- Store-level performance monitoring

SQL Analysis Performed

SQL was used for:

- Revenue aggregation
- Monthly sales trend analysis
- Product ranking
- Inventory risk calculation
- Revenue contribution analysis
- Customer segmentation
- Promotion performance evaluation
- Regional performance analysis

Business Value Delivered

- Identified major inventory gaps impacting revenue performance
- Highlighted high-demand products vulnerable to stockouts
- Improved visibility into operational supply chain risks
- Supported data-driven inventory optimization decisions
- Demonstrated how operational inefficiencies affect business performance

Future Enhancements

Potential future improvements include:

- Demand forecasting using Python (ARIMA / Prophet)
- Inventory optimization modeling
- Real-time inventory monitoring dashboard
- Automated stockout alert system
- Supply chain risk prediction models
- Streamlit or Power BI executive dashboard

Conclusion

Through this project, it is evident that the concepts of retail analytics and business intelligence play an important role in helping an organization detect areas of inefficiency, mitigate risks in the supply chain, and better manage its inventory.

Through sales analysis, inventory risk management, and business strategy, the project offers a way for organizations to improve their revenue performance.